

Core Glossary

Plain-language definitions for reference, not a memorization list

Prepared for Yasin Kaymaz, PhD | Version 1.0 | 14 September 2026

This is a teaching package, not a clinical, legal or institutional accreditation document. Scientific sources are identified by [Sxx] in the Course Source Register.

Use the word with its biological unit

These definitions support the specific course exercises. Source families: molecular foundations[S01]; formats[S03,S04]; statistics[S07-S10]; data/modeling[S11-S15]; structure[S16,S17]; variants[S18,S19]; privacy[S20-S22]. A specialist application may need additional definitions and current standards.

Term	Meaning
Allele	One sequence form at a locus; interpretation depends on the reference and biological context.
Annotation	Information about features such as genes and transcripts on a particular reference sequence.
Assay	A measurement procedure directed at a biological property or molecule.
Assembly	A reference representation of a genome sequence; coordinates depend on its version.
BED	A common interval format using zero-based starts and end-exclusive intervals.
BLAST	A sequence-search approach that finds local similarities using heuristic search steps.
Batch effect	Variation associated with a processing group or run rather than the biological question of interest.
Biological replicate	An independently sampled or assigned biological unit appropriate to the question.
Cell type	A biological identity inferred using multiple lines of evidence, not merely a cluster number.
Classifier	A model that predicts a label from input features.
Cluster	A grouping produced by a defined representation and algorithm; not automatically a biological entity.
Codon	A triplet of RNA bases read during translation under a specified genetic code.
Confounding	Two influences are entangled so their separate effects cannot be identified from the design alone.
Control	A comparison or procedure that helps interpret an experiment; it does not replace replication.
Count matrix	A table of measured or assigned counts, with explicitly labelled biological entities on its axes.
Coverage / depth	The amount of sequence evidence covering positions; mean depth does not describe every base.
DNA	A nucleic-acid sequence that stores heritable biological information.
Data leakage	Information unavailable at the intended prediction time influences model construction or evaluation.

Term	Meaning
Differential expression	A comparison of expression across conditions using a stated model and biological design.
Domain	A structurally or functionally meaningful region of a protein; boundaries and states require context.
Donor	The individual biological source of samples; multiple cells may be nested within one donor.
E-value	Expected count of chance alignments at least as strong as a score under a stated search model and space.
Effect size	The magnitude of an estimated difference or association, not its p-value.
Exon	A sequence retained in a processed transcript; not every exon base is translated.
FASTA	A text format for sequence identifiers and sequences.
FASTQ	A text format containing sequence reads and per-base quality information.
False discovery rate	Expected proportion of false rejections among a defined family of discoveries, with zero when none are made.
Feature	A measured or derived variable used in an analysis, such as a gene value.
Fine-tuning	Adapting a pretrained model using an additional task/data training process.
Fold change	A ratio between quantities, interpreted with a stated normalization and zero-handling rule.
Foundation model	A broadly pretrained model that can be adapted or evaluated on multiple tasks; usefulness is task-dependent.
Gene	A genomic region associated with a functional product, which can be RNA rather than protein.
Genome	The genetic sequence complement represented for an organism or biological entity.
Heuristic	A strategy designed for practical search efficiency without guaranteeing every exact optimum under all settings.
Hypothesis	A testable proposed explanation or statistical statement, not a proven fact.
Intron	A sequence removed during a specified RNA splicing process.
Isoform	An alternative transcript or protein form; specify which type is meant.
Library	Prepared molecular material for sequencing; not synonymous with an independent donor.
Local alignment	An alignment optimizing similarity for selected subsequences under a specified score.
Log2 fold change	The base-two logarithm of a stated ratio; +1 is doubling and -1 is halving.

Term	Meaning
Metadata	Information describing samples, conditions, processing and identifiers.
Normalization	A transformation intended to make specified quantities comparable; it does not solve every design defect.
Null model	A specified statistical model used as a reference for a test or simulation.
PAE	Predicted aligned error: a model estimate of positional error conditional on alignment to another part of a structure.
PCA	A linear representation summarizing directions of variation; plotted separation needs biological interpretation.
Precision of a claim	How specifically a conclusion is stated; do not imply greater certainty than evidence supports.
Protein	A polypeptide biological molecule whose function depends on sequence, context, structure and state.
Pseudobulk	Aggregation of cell counts within a biologically meaningful replicate and cell grouping, retaining replication.
Pseudonymization	Replacing/separating identifying information; it does not by itself establish anonymization.
RNA	A nucleic acid with many roles, including messenger, regulatory and structural functions.
Randomization	Assigning units or processing order by a declared chance procedure to reduce systematic bias.
Read pair	Two sequenced ends from a paired-end fragment; pairs and individual reads have different base budgets.
Reference	The specified sequence or dataset against which observations are described.
Reproducibility	Ability to recover results from the stated data and procedure; a reproducible method can still be wrong.
Runtime	The running computational session and its current memory/state.
SAM / BAM	Text/binary formats for alignments; text fields and binary/API coordinates can differ.
Seed	An input controlling a pseudo-random sequence for reproducibility, not proof of authorship.
Sensitivity	True positives divided by all actual positives for a specified classification task.
Single-cell profiling	Measurements associated with individual cells, with cells often nested in biological samples.
Specificity	True negatives divided by all actual negatives for a specified classification task.
Splicing	RNA processing that removes specified introns and joins retained sequences.

Term	Meaning
TPM	Transcripts per million, an abundance scale based on length-adjusted rates within a sample.
Technical replicate	A repeated measurement or preparation from the same biological material.
Test set	Data reserved to evaluate a locked analysis without influencing model selection.
Transcript	An RNA product associated with transcription and processing.
Transcription	Synthesis of RNA from a DNA template.
Translation	Synthesis of a polypeptide by reading a coding RNA sequence.
UMAP	A nonlinear neighborhood-oriented representation whose appearance depends on choices and data.
VCF	A format for variants relative to a reference, with metadata and fields requiring careful interpretation.
VUS	Variant of uncertain significance: evidence is insufficient for a definitive classification in the relevant context.
Validation	Testing a claim against appropriate evidence; the required evidence depends on the intended use.
Variant	A sequence difference relative to a specified reference representation.
Zero-shot evaluation	Evaluation without task-specific parameter adaptation in the declared setting; definitions and setup must be stated.
p-value	A tail probability calculated under a null model and assumptions, not the posterior probability that the null is true.
pLDDT	A local structural-confidence score, not affinity or a probability of clinical effect.